Campus Stay - Assessment Submission

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Project: Campus Stay - Student Accommodation
Repository: https://github.com/deen123123/comp3613_a2_816036619

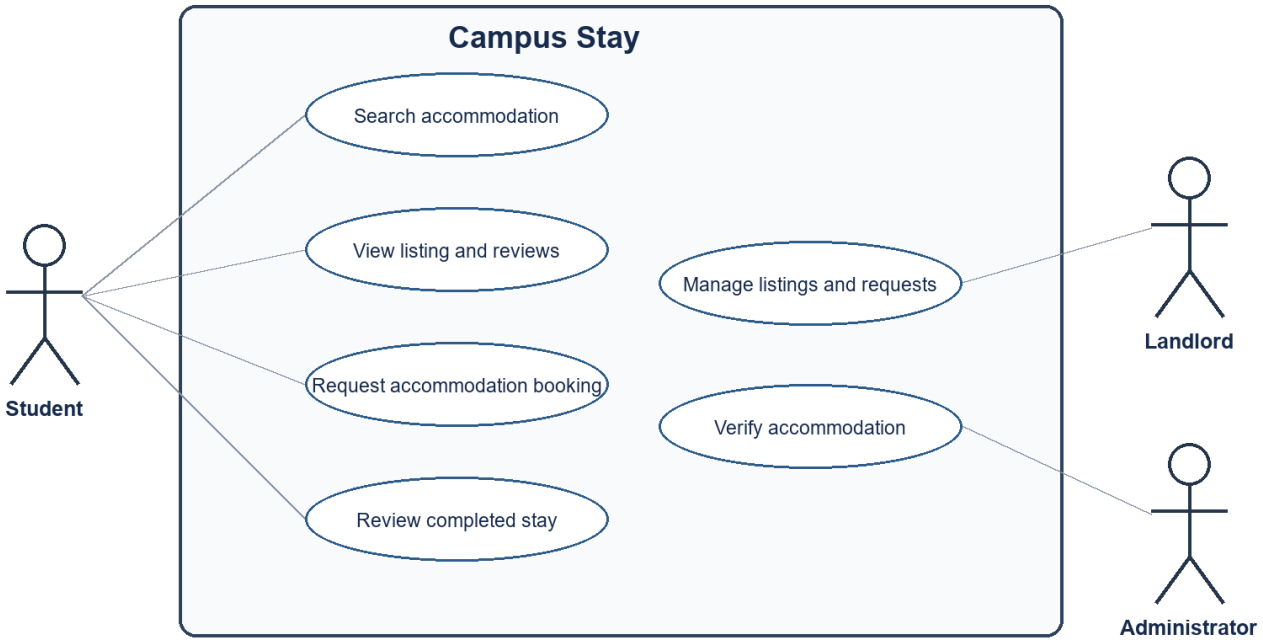
Project description

Campus Stay is a student accommodation marketplace for finding housing near the University of the West Indies. Students search verified listings, compare details and reviews, and request bookings. Landlords manage properties and respond to booking requests. Administrators verify new listings before public display.

Implementation summary

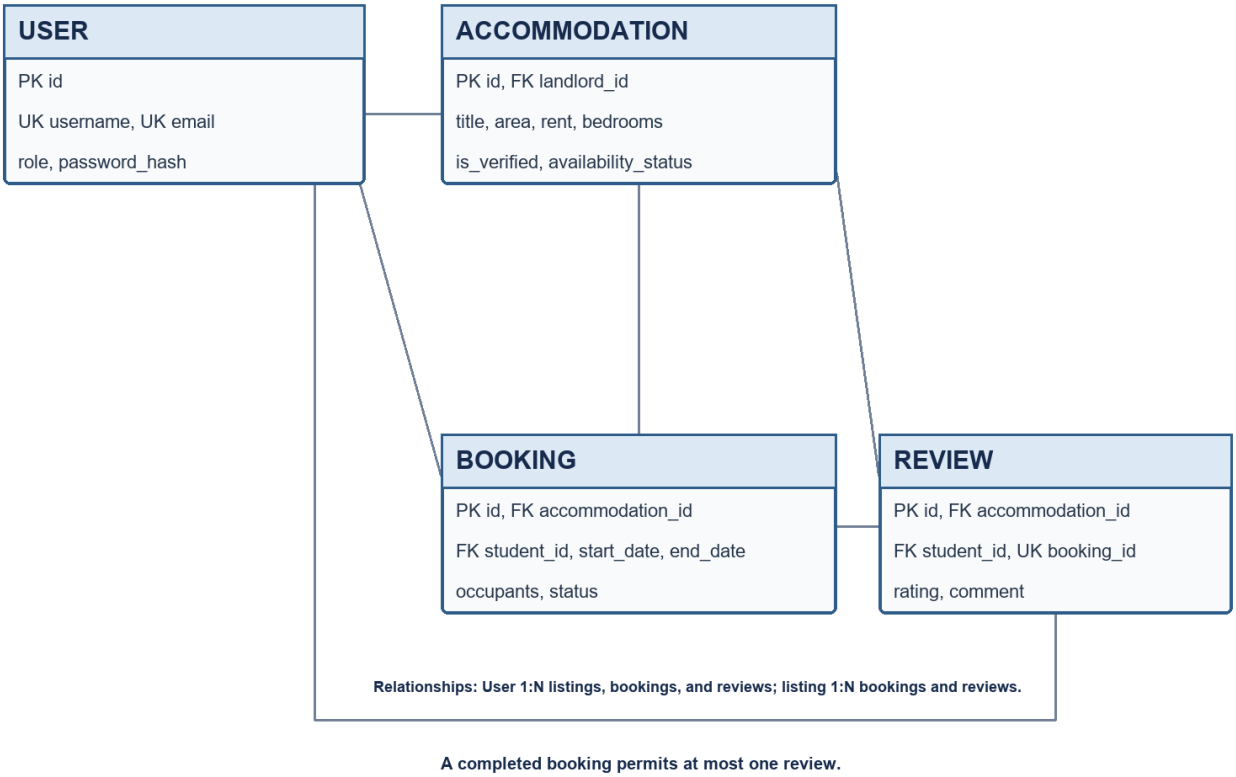
The application is implemented with FastAPI, SQLModel, Jinja templates, and SQLite for local development. Routes render the student interface and expose JSON endpoints. Business rules are organized in services and repositories. Session-based authorization separates student, landlord, and administrator actions. Public accommodation search exposes verified listings only.

Use case diagram



Students search, view, request bookings, and review completed stays. Landlords manage listings and booking requests; administrators verify accommodation.

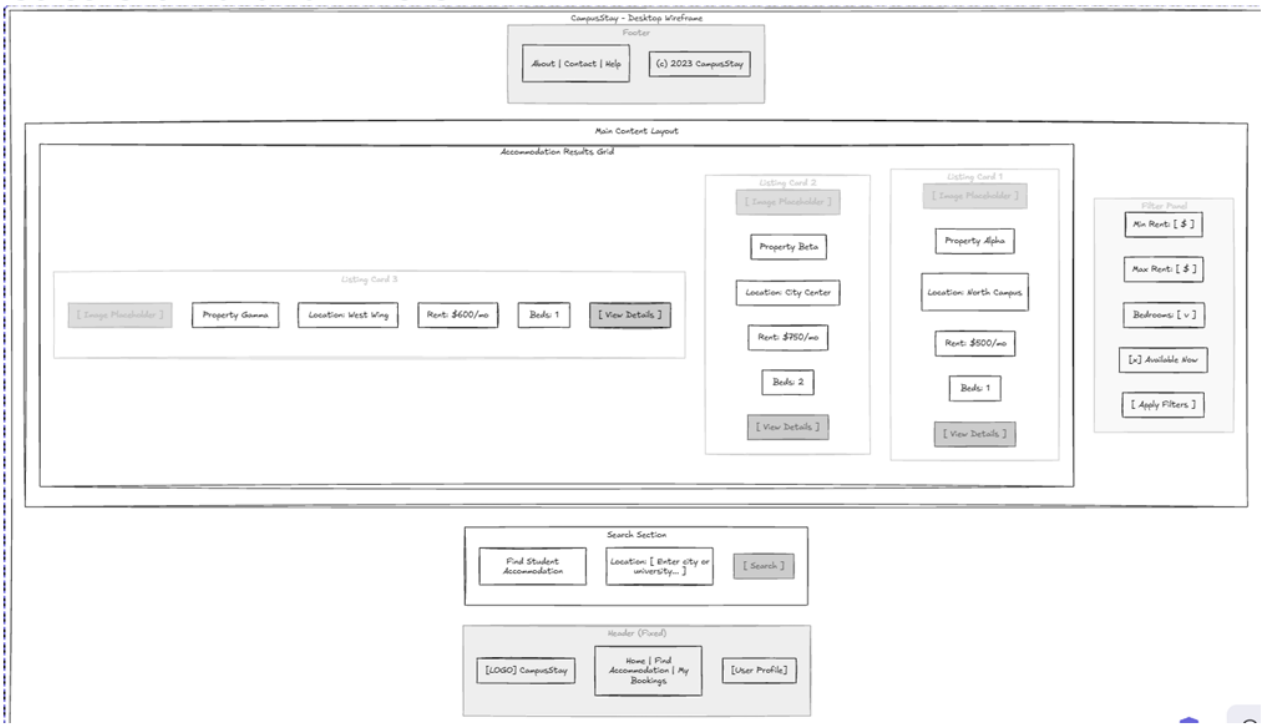
Entity Relationship Diagram



Data relationships

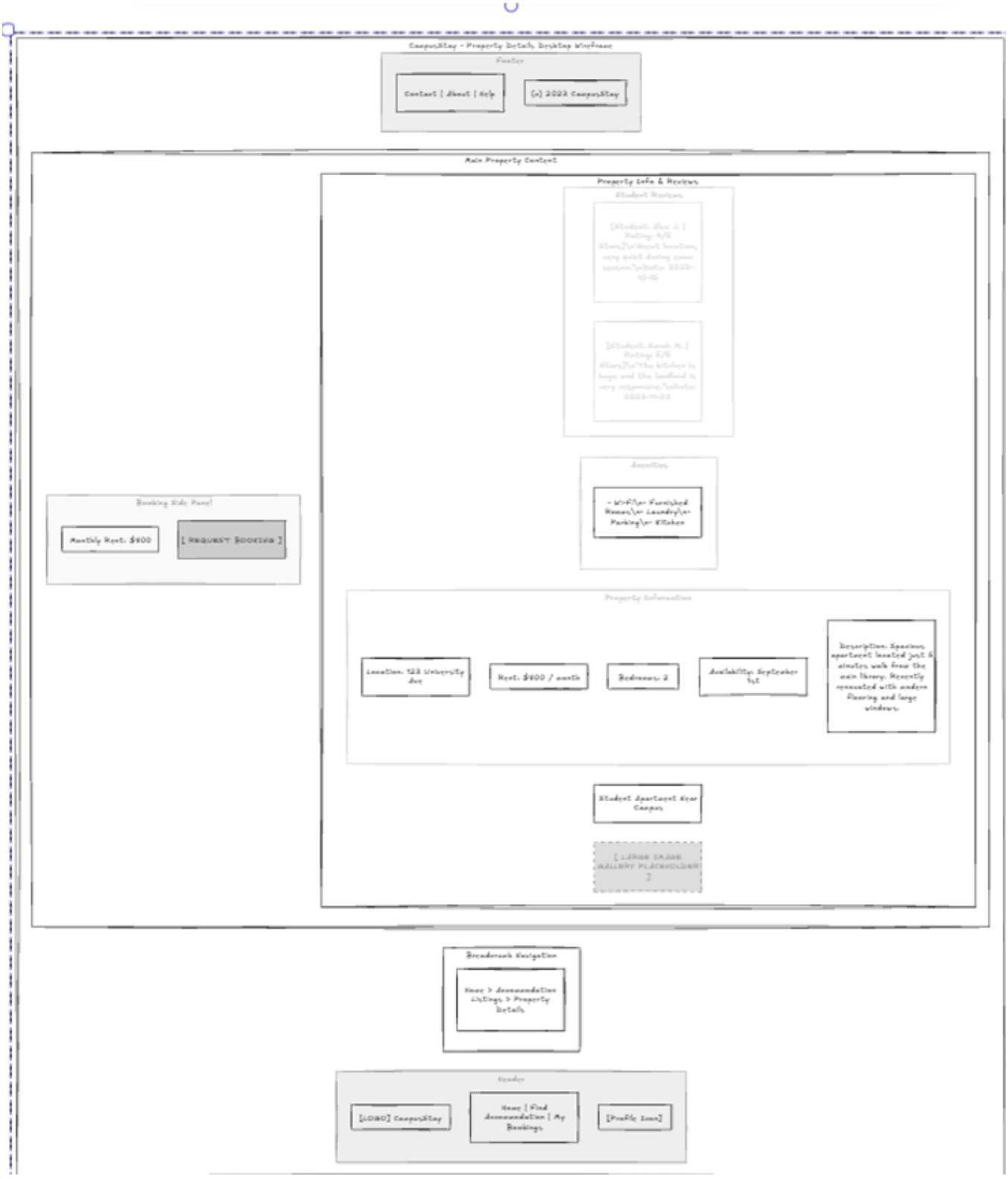
A user may own multiple accommodations, request multiple bookings, and write reviews. An accommodation receives bookings and reviews. A review links to a booking, so only a completed stay can be reviewed and a unique booking relationship prevents a second review for the same stay.

Application Wireframe - Search Results



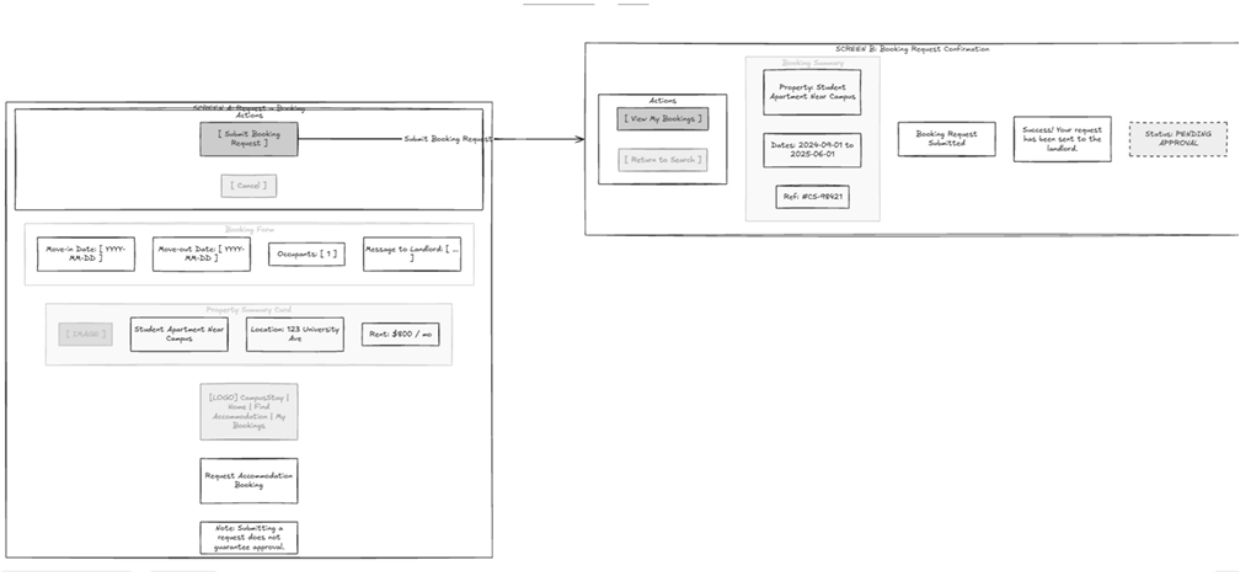
The student searches by location and refines results by rent, bedrooms, and current availability.

Application Wireframe - Listing Details and Reviews



The student checks property information, rent, amenities, availability, and review content.

Application Wireframe - Booking Request and Confirmation



The student enters dates, occupants, and an optional message, then receives a pending booking confirmation.

Test Plans

Test environment and execution

Python 3.10+ and project requirements. Tests use pytest, FastAPI TestClient, and a temporary in-memory SQLite database with seeded fixtures. Run from the repository root with: python -m pytest -v. Unit-only: python -m pytest -v tests/unit. Integration: python -m pytest -v tests/integration tests/test_workflows.py.

Unit test plan

U-01 clean optional text; U-02 integer parsing and inclusive bounds; U-03 malformed and out-of-range integers; U-04 floating-point validation; U-05 ISO date parsing; U-06 optional integer conversion; U-07 accommodation search parameter normalization; U-08 safe defaults for invalid filters and sort; U-09 availability checkbox parsing; U-10 booking reference and checkout-exclusive stay duration; U-11 pending, active, completed, and declined booking display status.

Integration test plan

I-01 health check response; I-02 location and rent search; I-03 available-now filtering; I-04 exclusion of unverified listings; I-05 JSON card response contract; I-06 listing detail and reviews; I-07 booking validation for date, overlap, and availability rules; I-08 booking confirmation and landlord decision; I-09 role-based access; I-10 overlap conflict handling; I-11 completed-stay review eligibility; I-12 admin verification controls public visibility; I-13 landlord-created listing begins unverified.

API test plan

Postman collection requests: GET /health; GET /api/accommodations; location and rent filtering for Arima; available_now filtering; and a Valsayn query that confirms unverified accommodation is hidden. Each request includes Postman test scripts for status, response shape, and expected filter behavior.

User Acceptance Test Plan

Tester: Student / intended student user. Preconditions: application is running with seeded data and the tester signs in as bob / bobpass. Mark each case Pass only when all success criteria are observed; record defects and date.

UAT-01 - Search accommodation

Using the search-results wireframe, search Curepe and apply a maximum rent of TT$2,000. Clear filters, select Bedrooms 2 and Available Now, and apply. Pass when the results reflect the requested criteria and show title, location, monthly rent, bedrooms, and a View Details action.

UAT-02 - View listing and reviews

Open View Details for the verified Sunny private room. Pass when the listing details, monthly rent, availability, and verified-stay reviews correspond to the selected property and an authenticated student can proceed to booking when available.

UAT-03 - Request accommodation booking

Enter valid future move-in and move-out dates at least 28 days apart, occupants within the listing limit, and an optional message. Pass when one request is created and confirmation shows the correct property and dates, a booking reference, and Pending Approval status. Invalid or unavailable requests must not be accepted.

UAT execution and sign-off

UAT-01 Search: Pending student demonstration.
UAT-02 Listing and reviews: Pending student demonstration.
UAT-03 Booking request: Pending student demonstration.
Overall acceptance: Pending.
Tester name/signature: _____ Date: _____

Unit Test Report

Command: python -m pytest -v tests/unit

Result: PASS - 19 tests passed. Image below is rendered from output captured during the actual pytest run.

Unit test run - actual pytest output

```
============================= test session starts =============================
platform win32 -- Python 3.14.2, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\dines\AppData\Local\Programs\Python\Python314\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\dines\Downloads\Campus-Stay-COMP-3613-Assignment-1--main
configfile: pyproject.toml
plugins: anyio-4.12.1, asyncio-1.4.0
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collecting ... collected 19 items

tests/unit/test_accommodation_filters.py::test_filters_normalize_search_and_filter_parameters PASSED [  5%]
tests/unit/test_accommodation_filters.py::test_filters_use_safe_defaults_for_unknown_sort_and_invalid_numbers PASSED [ 10%]
tests/unit/test_accommodation_filters.py::test_filters_accept_supported_true_values PASSED [ 15%]
tests/unit/test_booking_model.py::test_booking_reference_is_derived_from_persisted_id PASSED [ 21%]
tests/unit/test_booking_model.py::test_booking_nights_uses_checkout_exclusive_duration PASSED [ 26%]
tests/unit/test_booking_model.py::test_display_status_reflects_booking_state_and_dates[pending--5-5-pending] PASSED [ 31%]
tests/unit/test_booking_model.py::test_display_status_reflects_booking_state_and_dates[approved--5-10-active] PASSED [ 36%]
tests/unit/test_booking_model.py::test_display_status_reflects_booking_state_and_dates[approved--10--1-completed] PASSED [ 42%]
tests/unit/test_booking_model.py::test_display_status_reflects_booking_state_and_dates[declined--10--1-declined] PASSED [ 47%]
tests/unit/test_validators.py::test_clean_trims_optional_text PASSED     [ 52%]
tests/unit/test_validators.py::test_parse_int_accepts_values_in_range[1-1-10-1] PASSED [ 57%]
tests/unit/test_validators.py::test_parse_int_accepts_values_in_range[10-1-10-10] PASSED [ 63%]
tests/unit/test_validators.py::test_parse_int_accepts_values_in_range[4-None-5-4] PASSED [ 68%]
tests/unit/test_validators.py::test_parse_int_rejects_invalid_or_out_of_range_values[abc-1-10-whole number] PASSED [ 73%]
tests/unit/test_validators.py::test_parse_int_rejects_invalid_or_out_of_range_values[0-1-10-at least 1] PASSED [ 78%]
tests/unit/test_validators.py::test_parse_int_rejects_invalid_or_out_of_range_values[11-1-10-at most 10] PASSED [ 84%]
tests/unit/test_validators.py::test_parse_float_enforces_numeric_range PASSED [ 89%]
tests/unit/test_validators.py::test_parse_date_accepts_iso_format_and_rejects_other_formats PASSED [ 94%]
tests/unit/test_validators.py::test_to_int_or_none_returns_none_for_blank_or_non_numeric_values PASSED [100%]

============================= 19 passed in 0.09s ==============================
```

Integration Test Report

Command: python -m pytest -v tests/integration tests/test_workflows.py
Result: PASS - 19 tests passed. Full suite: 38 passed. Image below is rendered from output captured during the actual pytest run.

Integration test run - actual pytest output

```
============================= test session starts =============================
platform win32 -- Python 3.14.2, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\dines\AppData\Local\Programs\Python\Python314\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\dines\Downloads\Campus-Stay-COMP-3613-Assignment-1--main
configfile: pyproject.toml
plugins: anyio-4.12.1, asyncio-1.4.0
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collecting ... collected 19 items

tests/integration/test_api.py::test_health_endpoint_returns_success_status PASSED [  5%]
tests/integration/test_api.py::test_accommodation_api_returns_verified_listings_and_card_fields PASSED [ 10%]
tests/integration/test_api.py::test_accommodation_api_applies_location_and_rent_filters PASSED [ 15%]
tests/integration/test_api.py::test_accommodation_api_available_now_filter_excludes_unavailable_listings PASSED [ 21%]
tests/integration/test_api.py::test_accommodation_api_does_not_expose_unverified_listings PASSED [ 26%]
tests/test_workflows.py::test_health PASSED                              [ 31%]
tests/test_workflows.py::test_login_and_role_redirects PASSED            [ 36%]
tests/test_workflows.py::test_protected_pages_redirect_to_login PASSED   [ 42%]
tests/test_workflows.py::test_search_by_location_then_filters PASSED     [ 47%]
tests/test_workflows.py::test_available_now_filter_hides_unavailable PASSED [ 52%]
tests/test_workflows.py::test_api_accommodations PASSED                  [ 57%]
tests/test_workflows.py::test_detail_shows_reviews_and_rent PASSED       [ 63%]
tests/test_workflows.py::test_booking_rules_and_landlord_decision PASSED [ 68%]
tests/test_workflows.py::test_cannot_book_unavailable_accommodation PASSED [ 73%]
tests/test_workflows.py::test_approving_closes_overlapping_requests PASSED [ 78%]
tests/test_workflows.py::test_other_landlords_cannot_decide PASSED       [ 84%]
tests/test_workflows.py::test_review_only_after_stay_and_only_once PASSED [ 89%]
tests/test_workflows.py::test_admin_verification_controls_visibility PASSED [ 94%]
tests/test_workflows.py::test_landlord_accommodation_starts_unverified PASSED [100%]

============================= 19 passed in 8.91s ==============================
```

API Tests - Postman Collection and Run Report

Exported collection: postman/Campus-Stay-API.postma
n_collection.json (Postman Collection v2.1). Set baseUrl to http://127.0.0.1:5000, initialize demo data with python manage.py init, start the app with python manage.py run, import the JSON into Postman, then run all requests in Collection Runner.

Collection test coverage

A-01 Health endpoint returns HTTP 200 and healthy JSON.
A-02 Accommodation response is a JSON array with required fields and verified listings.
A-03 Arima location and rent range filters return matching listings.
A-04 Available-now filter excludes unavailable Trincity.
A-05 Unverified Valsayn listing is not returned publicly.

Postman Collection Runner report screenshot

PENDING - Postman/Newman was not available in this workspace, so the collection has not been run here. This report does not claim a Postman pass. Run the exported collection in Postman and insert the actual Collection Runner screenshot before final submission.

Overall evidence status

Unit tests: PASS (19). Integration tests: PASS (19). Full pytest suite: PASS (38). Postman Runner screenshot: pending actual Postman run. UAT execution and sign-off: pending student demonstration.